Machine-verified rooted-tree majorants for polymer expansions with holes

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Abstract

We present a self-contained, fully machine-verified quantitative toolkit for cluster expansions of polymer systems with excluded regions (*holes*), in the discrete geometry of Dimock's renormalization-group framework. Three Lean 4 theorems, compiled against a pinned Mathlib revision with a clean axiom oracle, provide: (i) a normalized child-factorial bound over spanning trees of the complete graph (the rooted-tree majorant $\leq 4^n$); (ii) a marked-root leaf summation for the second Ursell expansion with holes, with the moment constant paid once at the root and closed leaf ratio $4M^2$ per additional vertex; and (iii) a target-preserving orderwise bound in which the exact target union survives until the modified-metric exponential is extracted. A certified companion (interval arithmetic, 120-bit precision, committed transcript) tabulates the admissible smallness gate and encloses every derived constant, including the threshold $\varepsilon^* = \Lambda^{-1}$ at which downstream activity series with per-vertex factor ε have geometric ratio below one (the Lean endpoints prove the coefficient bounds $M\Lambda^n$; no activity-series convergence is claimed as formalized). Non-vacuity is machine-checked: a concrete hole family satisfying every hypothesis is exhibited in Lean and both polymer-facing endpoints are instantiated at it with a strictly positive weight. No claim beyond the proved statements is made: this is expansion tooling with machine constants, originating from (but logically independent of) a lattice Yang–Mills formalization programme.

1 Introduction and honest scope

Cluster (polymer) expansions convert log-partition functions into sums over connected geometric objects; their convergence proofs hinge on combinatorial majorants for tree sums and on geometric smallness gates. When the expansion must respect *holes* — excluded regions that the expansion sees as atomic units, as in renormalization-group treatments of large-field regions — the bookkeeping is notoriously error-prone: root normalizations, child factorials, and hole-separation hypotheses interact in ways that have historically produced wrong constants in the literature[1].

This paper's contribution is deliberately narrow and fully witnessed:

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*With machine collaboration; every theorem below is machine-checked in Lean 4 against a pinned Mathlib, and every numerical constant is enclosed by committed interval arithmetic. Repository: https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME.

[1] Our own development measured exactly such an error in its pre-registered prediction of the smallness-gate crossover; see Remark 4.1. The correction protocol — constants are computed, never assembled by hand — is part of the method.

- **Exact** (Lean theorems, pinned toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned by commit): Theorems 3.1, 3.3, 3.4, and the non-vacuity instantiations of Section 5.
- **Certified** (committed interval-arithmetic transcript, `python-flint`/Arb, 120-bit precision, ball-plus-boolean on every enclosure): the admissibility table of Section 4.
- No Clay-adjacent or continuum claim of any kind appears. Importance is not inherited by thematic proximity; only a proved reduction transfers it, and none is claimed here.

[SLOT: oracle output of the three endpoints post-build; axiom list must print exactly [propext, Classical.choice, Quot.sound]]

2 The discrete geometry

Definition 2.1 (cube lattice and adjacency)**.** *For $d, L \in \mathbb{N}$, a* cube *is a point of $(\mathbb{Z}/L)^d$ (Lean: `Cube d L := Fin d → ZMod L`). Two cubes are adjacent (`cubeAdj`) iff they are distinct and each coordinate differs by 0 or ± 1: the sup-metric neighbor graph, each cube having $3^d - 1$ neighbors.*

Definition 2.2 (hole family)**.** *A* hole family *(Lean: `HoleFamily d L`) is a finite set of finite sets of cubes — the connected components of the excluded region. A polymer X* respects *the family iff every hole is contained in X or disjoint from it. The bounds below hypothesize: pairwise disjointness of holes, no `cubeAdj` edge between distinct holes, and non-emptiness of each hole.*

Definition 2.3 (weights and constants)**.** *For a rate κ, the hole weight of a polymer X is $e^{-\kappa\,(\rho(X)+1)}$ with ρ the discrete modified metric (Lean: `appendixFHoleExpWeight`). The smallness gate at dimension d and rate κ_0 is*

$$G(d, \kappa_0) \;=\; (3^d)^2\, e^{-\kappa_0}\, 2^{3^d+1} \;<\; 1 \qquad (\textit{Lean hypothesis } h_{Cq}), \tag{1}$$

and the derived constants are

$$\mathrm{RS} = (1-G)^{-1}, \qquad M = \max\!\big(1,\; \kappa_0^{-1},\; \mathrm{RS}\big), \qquad \Lambda = 4M^2, \tag{2}$$

(Lean: `appendixFSecondUrsellMomentConstant`, `appendixFSecondUrsellLeafConstant`). In applications where the order-n term additionally carries an activity factor ε^n, the resulting geometric series has ratio $\Lambda\varepsilon$, which is below one exactly for $\varepsilon < \varepsilon^* := \Lambda^{-1}$ (strict; nothing is claimed at equality). We emphasize the tricotomy status: the Lean endpoints prove the coefficient bounds $M\Lambda^n$ (exact)*; the enclosures of Λ^{-1} are* certified*; the passage from coefficient bound to a convergent activity series is paper-level algebra conditional on the stated ε^n structure — no activity-weighted series is formalized in the cited endpoints.*

3 The three endpoints (exact)

All three are stable aliases in module `MarkedRootedClosure` (file `MarkedRootedClosure/PaperTheorems.lean`); each is an exact application of a pinned upstream theorem — no new axiom, no opaque assumption.

Theorem 3.1 (rooted-tree majorant; `normalizedRootedChildFactorialTreeBound`). *For every $n \in \mathbb{N}$,*

$$\frac{n+1}{(n+1)!} \sum_{T \in \mathcal{T}(K_{n+1})} \prod_{v} \big(c_T(v)\big)! \;\le\; 4^n,$$

where $\mathcal{T}(K_{n+1})$ is the set of spanning trees of the complete graph on $n+1$ vertices and $c_T(v)$ the child count of v in T rooted at the marked vertex.

Remark 3.2 (exact value — upgrade slot). *The left-hand sum satisfies the* exact identity $\sum_T \prod_v c_T(v)! = n!\, C_n$ *with C_n the nth Catalan number, machine-proved in the companion repository* ***`rooted-tree-catalan-closure`****; since $(n+1)\, n!\, C_n/(n+1)! = C_n \le 4^n$, Theorem 3.1 is its corollary, and the majorant overcounts by only the factor $4^n/C_n \sim \sqrt{\pi}\, n^{3/2}$.* ***[SLOT: the identity becomes a theorem OF THIS PAPER only when its module compiles green in the pinned core of this repository (merge tracked under judge J-C1-1); until then it is cited, not claimed]***

Theorem 3.3 (marked-root leaf summation; `markedRootLeafGeometricBound`). *Let HF be a hole family in dimension d with the separation hypotheses above, $w \ge 0$ a polymer weight dominated by the hole weight at rate $2\kappa_0$, r a marked root cube, and suppose the gate (1). Then for every n,*

$$(n+1) \cdot S^{\#}_{HF,w,r}(n) \;\le\; M \cdot \Lambda^n,$$

where $S^{\#}$ is the weighted marked-root tree sum of the second Ursell expansion (Lean: `appendixFHoleHsharpWeigh` and M, Λ are the constants of (2): the moment constant is paid once *at the root; each additional vertex costs exactly the closed leaf ratio $\Lambda = 4M^2$.*

Theorem 3.4 (target-preserving orderwise bound; `targetPreservingWeightedTreeBound`). *With the same geometric hypotheses, let $w \le e^{-\mathrm{rate}\,(\rho+1)} \cdot u$ be a split of the weight into an extracted exponential at a nonnegative rate times a remainder u dominated by the hole weight at rate $2\kappa_0$, and let Y be a target union whose skeleton contains a root. Then*

$$T^{\#}_{HF,w,Y}(n) \;\le\; M \cdot e^{-\mathrm{rate}\,(\rho(Y)+1)} \cdot \Lambda^n.$$

The exact target union remains present until the modified-metric exponential has been extracted; only then is the marked-root estimate applied — the orderwise bookkeeping that hole-respecting expansions require.

4 The certified admissibility companion

Script `scripts/c1_admissibility_arb.py` (sha256 `f114fc4b…`, `python-flint` 0.9.0, precision 120 bits; committed transcript `scripts/c1_admissibility_transcript.txt`, exit 0) encloses G, M, Λ, ε^* with ball-plus-boolean output; the gate booleans are provable-only (PASS printed only when $G < 1$ holds for every point of the enclosing balls). Selected rows (midpoints shown; radii in the transcript, all below 3×10^{-6} relative):

d	κ_0	gate G	M	Λ	ε^*
2	11	1.3853059	FAIL (provably ≥ 1)		
2	11.5	0.84023048	6.2590162	156.70114	0.0063815746
2	12	0.50962555	2.0392580	16.634292	0.060116775
2	**16**	0.0093341175	1.0094221	4.0757316	**0.24535472**
2	24	3.13×10^{-6}	1.0000031	4.0000251	0.24999843
3	25	2.7177241	FAIL (provably ≥ 1)		
3	26	0.99979481	4873.5020	9.50×10^7	1.05×10^{-8}
3	28	0.13530751	1.1564805	5.3497886	0.18692327
3	**33**	0.00091169486	1.0009125	4.0073035	**0.24954436**

The certified crossovers of the gate are

$$\kappa^*(2) \in 11.32592096 \pm 2.72 \times 10^{-10}, \qquad \kappa^*(3) \in 25.99979479 \pm 2.32 \times 10^{-9},$$

and $\varepsilon^* \uparrow 1/4$ as $\kappa_0 \to \infty$ (the pure tree-majorant limit $\Lambda \downarrow 4$). The row $(d, \kappa_0) = (3, 26)$ sits just above $\kappa^*(3)$ and displays the sharpness of the gate: admissible, but with $M \approx 4.9 \times 10^3$ and $\varepsilon^* \approx 1.05 \times 10^{-8}$.

Scope of the certificates: the transcript's band booleans certify the gate at the two registered band endpoints $\kappa_0 = 16$ and $\kappa_0 = 33$; the extension to all larger κ_0 is the exact monotonicity of $\kappa_0 \mapsto e^{-\kappa_0}$ (one line, not part of the certified output). Admissibility everywhere means the strict gate $G < 1$, as in the Lean hypothesis.

Remark 4.1 (a measured registration failure)**.** *The pre-registered charter predicted crossovers* 15.72 *and* 32.60 *from a hand-assembled expression carrying* $(3^d)^4$ *where the gate* (1) *carries* $(3^d)^2$*. The certified run measured the discrepancy (transcript, "crossover audit":* REGISTERED PREDICTION DOES NOT RE-DERIVE*); the registered admissibility bands* $\kappa_0 \geq 16$ *(d=2) and* $\kappa_0 \geq 33$ *(d=3) remain true — sufficient, not sharp — and only certified values are inked here. We record the failure because the method requires it: constants are computed, never assembled.*

5 Non-vacuity, machine-checked

The vacuity failure mode of formalized expansion bounds is hypotheses that no concrete geometry satisfies. Module `MarkedRootedClosure/NonVacuity.lean` exhibits — as instance terms, not existentials — the hole family HF^w in $d = 2$, $L = 8$ with two singleton holes at sup-distance 4, and proves by kernel decision (`decide`) that the holes are disjoint, edge-separated, and nonempty; the gate (1) at $\kappa_0 = 16$ is proved in Lean via $e^{16} > 2.7^{16} > 82944$ — the Lean-exact counterpart of the certified row $G \approx 0.00933$. Both polymer endpoints are then instantiated at HF^w with the *strictly positive* weight $e^{-32(\rho+1)}$ (no zero-weight degeneracy): `nonvacuous_markedRootLeafGeometricBound` and `nonvacuous_targetPreservingWeightedTreeBound`.

6 Provenance

- Lean toolchain `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit `0764272048…` (lakefile and manifest agree). **[SLOT: core job count and green-build record]**

- Certified companion: script sha256 `f114fc4bcf2193c47603fb595b91b4e3ea62addc3a1bfebe0a-a0c147c447f1f9`; transcript committed at the same revision as the script.
- Endpoint sources: `MarkedRootedClosure/` at repository root; upstream theorems in `YangMills/KP/` and `YangMills/RG/` of the same repository.
- The exact Catalan identity of Remark 3.2: repository `rooted-tree-catalan-closure` (committed axiom oracle log and CI replay); its Mathlib-facing `PlaneTree` companion is submitted upstream separately.